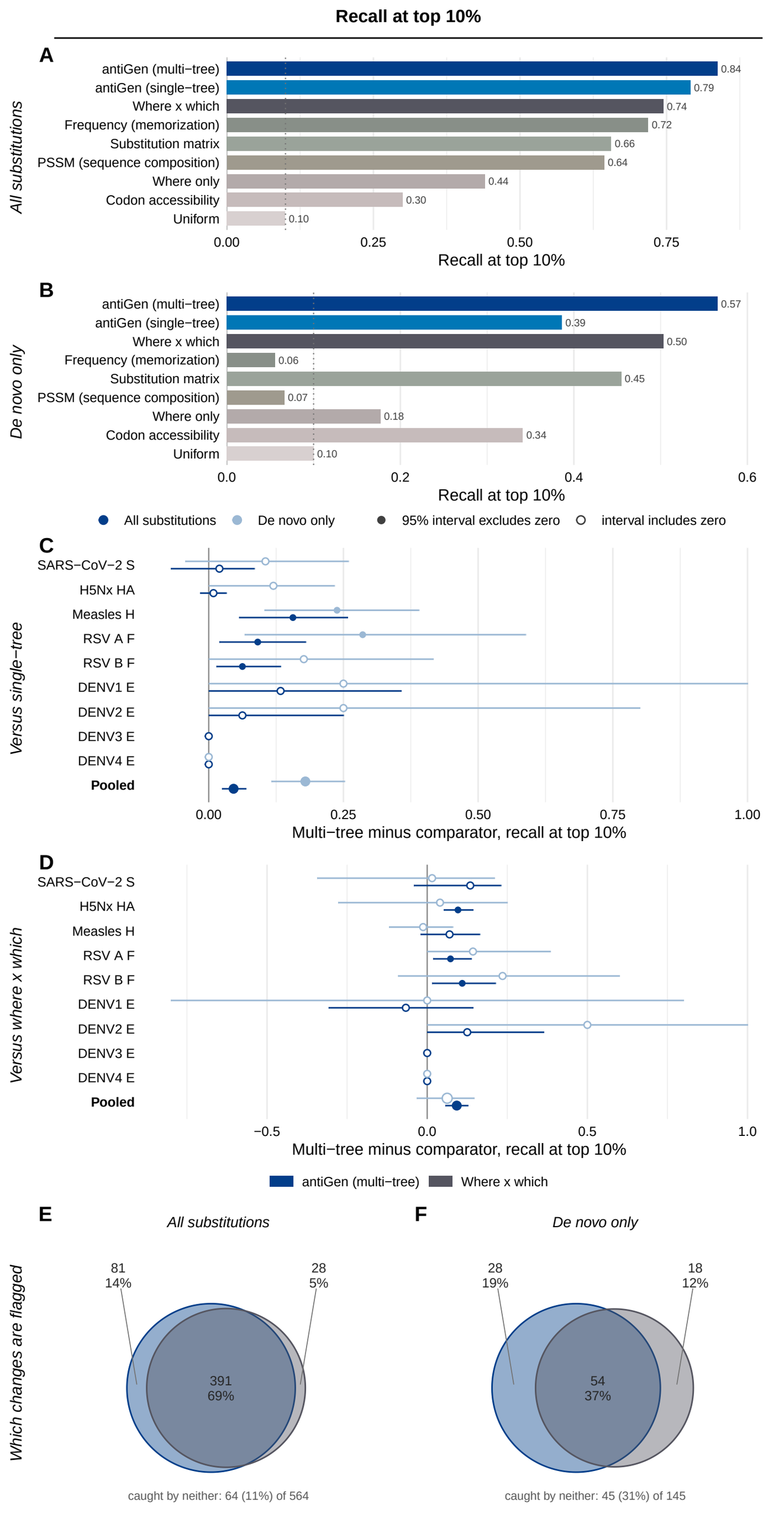
Recall at top 10%
A
All substitutions
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.84
0.79
0.74
0.72
0.66
0.64
0.44
0.30
0.10
0.00
0.25
0.50
0.75
Recall at top 10%
B
De novo only
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.57
0.39
0.50
0.06
0.45
0.07
0.18
0.34
0.10
0.0
0.2
0.4
0.6
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
0.00
0.25
0.50
0.75
1.00
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
antiGen (multi−tree)
Where x which
E
All substitutions
Which changes are flagged
81
14%
28
5%
391
69%
caught by neither: 64 (11%) of 564
F
De novo only
28
19%
18
12%
54
37%
caught by neither: 45 (31%) of 145